Lecture 12 (section 6.1)

The quantum search algorithm

- The classical algorithm to identify an item in an unstructured set of size N is of order

$$N$$

- The corresponding Grover's quantum algorithm requires

$$\sqrt{N}$$

operations.

$\Longrightarrow$ <u>The task</u>:

- The search space contains $N = 2^n$ elements; each element is indexed with an integer

$$x \in [0, N-1]\,, \qquad |x\rangle = \underbrace{|x_1 \ldots x_n\rangle}_{\text{corresponding binary reps}} \quad , \qquad x = 2^{n-1} \cdot x_1 + \cdots 2^0 \cdot x_n$$

- The search problem has M solutions; we typically assume that $M \ll N$.

- There is an *oracle* $\mathcal{O}$ — black box, performs unitary computation (we don't need to know) and tells us whether give x is a solution, *i.e.,* oracle computes

$$f(x) = \begin{cases} 1\,, & x \text{ is a solution} \\ 0\,, & x \text{ is not a solution} \end{cases}$$

- Let $|q\rangle$ be a single oracle bit:

$$|x\rangle\,|q\rangle \;\xrightarrow{\mathcal{O}}\; |x\rangle\,|q \oplus f(x)\rangle$$

i.e., , if $|q\rangle$ is a computational basis state, oracle flips it if x is a solution

- It is convenient to set $|q\rangle = |-\rangle$:

$$|x\rangle\,|q\rangle \;\xrightarrow{\mathcal{O}}\; \mathcal{O}\left[|x\rangle\,\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right] = \begin{cases} |x\rangle\,\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle)\,, & x \text{ a solution} \\ |x\rangle\,\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\,, & x \text{ not a solution} \end{cases}$$

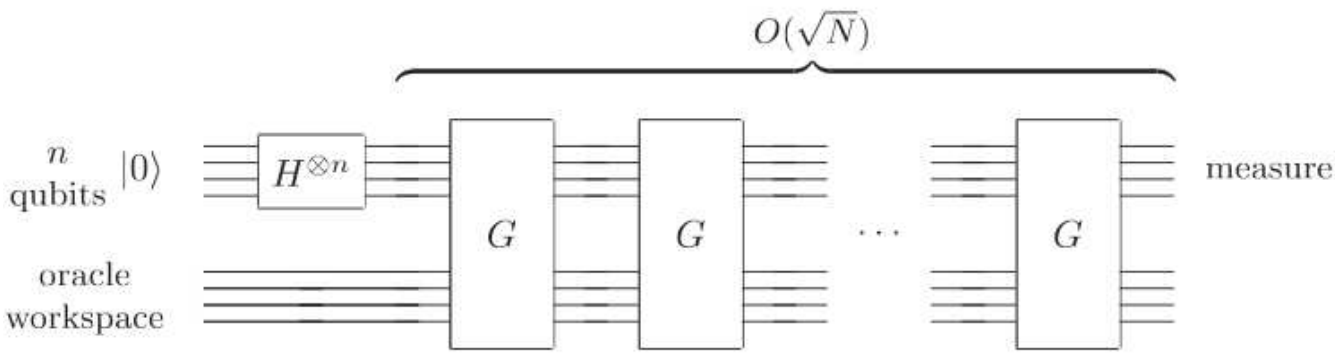
$$= |x\rangle\,(-1)^{f(x)}\,|-\rangle = (-1)^{f(x)}\,|x\rangle \otimes |q\rangle$$

Since oracle does not affect $|q\rangle$ in this setting, we assume that

$$\mathcal{O}\,|x\rangle = (-1)^{f(x)}\,|x\rangle$$

$\Longrightarrow$ <u>The procedure</u>:

- The full circuit:



  - step-1:

$$|0\rangle^{\otimes n} \;\xrightarrow{H^{\otimes n}}\; (H\,|0\rangle)^{\otimes n} = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)^{\otimes n} = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle \equiv |\psi\rangle\;, \qquad N = 2^n$$

where we called the equal superposition state above $|\psi\rangle$

  - step-2: apply the *Grover operator* G R-times,

$$|\psi\rangle \;\xrightarrow{G^R}\; G^R\,|\psi\rangle$$

where the integer R is given by

$$R = CI\left[\frac{\pi}{4}\cdot\sqrt{\frac{N}{M}}\right]$$

$CI(x)$ is *closest integer to* x, with convention that $CI(1.5) = 1$

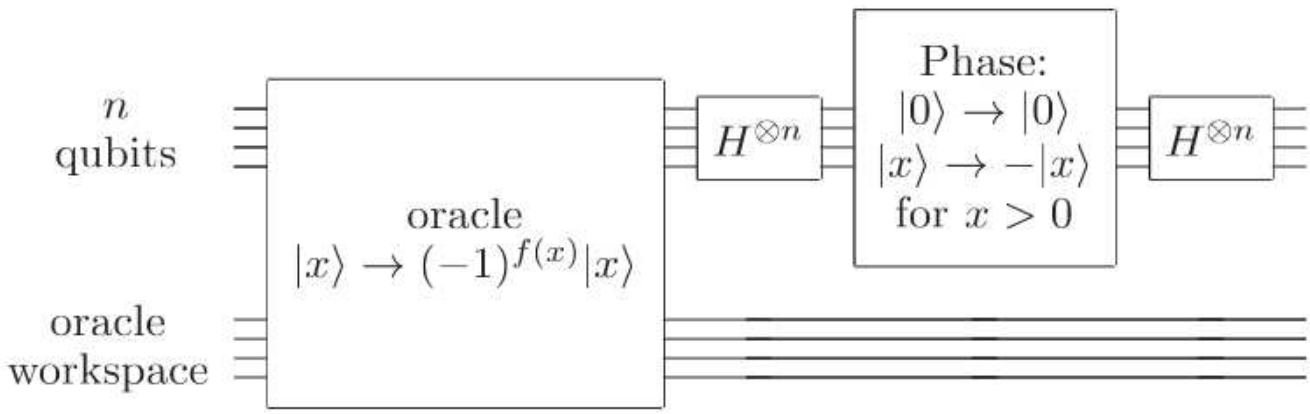
- step-3: measuring the result in the computational basis:

$$G^R |\psi\rangle \xrightarrow{\boxed{\nearrow}} \boxed{\nearrow}\, G^R |\psi\rangle = x^* \,,$$

where $f(x^*) = 1$ has an error of order $\frac{M}{N} \ll 1$

$\Longrightarrow$ <u>The Grover operator</u>:

- the circuit implementing G is given by:



- step-1: apply oracle to $|x\rangle$

$$\mathcal{O}\,|x\rangle = (-1)^{f(x)}\,|x\rangle$$

- The *Phase* flips all components of the state other than $|0\rangle \Longrightarrow$

$$Phase: \qquad |0\rangle\langle 0| - \sum_{x\neq 0} |x\rangle\langle x|$$

$$= 2\,|0\rangle\langle 0| - |0\rangle\langle 0| - \sum_{x\neq 0} |x\rangle\langle x| = 2\,|0\rangle\langle 0| - \underbrace{\sum_{x} |x\rangle\langle x|}_{=I_n} = 2\,|0\rangle\langle 0| - I_n$$

Remember that $|0\rangle$ actually has to be written in a binary notation:

$$2|0\rangle^{\otimes n}\langle 0|^{\otimes n} - I_n$$

- steps-2-3-4:

$$H^{\otimes n} \cdot Phase \cdot H^{\otimes n} = H^{\otimes n} \cdot \left(2|0\rangle^{\otimes n}\langle 0|^{\otimes n} - I_n\right) \cdot H^{\otimes n}$$

$$= 2 \cdot \underbrace{(H\,|0\rangle)^{\otimes n}}_{=|\psi\rangle} \; \underbrace{(\langle 0|\,H)^{\otimes n}}_{=\langle\psi|} \; - \; \underbrace{H^{\otimes n} \cdot H^{\otimes n}}_{=(H^2)^{\otimes n}=I_n} = 2 \cdot |\psi\rangle\,\langle\psi| - I_n$$

- to summarize, the full Grover iteration is

$$G \;\equiv\; \left(2 \cdot |\psi\rangle\,\langle\psi| - I_n\right) \cdot \mathcal{O}$$

$\Longrightarrow$ <u>What does the Grover iteration do?</u>

- define an equal superposition of solutions to the search problem β via $\sum'$, and the equal superposition of states that are not the solution α via $\sum''$:

$$|\alpha\rangle \equiv \frac{1}{\sqrt{N-M}} \sum_x{}'' \; |x\rangle \;, \qquad |\beta\rangle \equiv \frac{1}{\sqrt{M}} \sum_x{}' \; |x\rangle$$

Obviously,

$$\langle\alpha|\beta\rangle = 0$$

- Note that $|\psi\rangle$ is a superposition of span$\{|\alpha\rangle\,, |\beta\rangle\}$:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle = \frac{1}{\sqrt{N}}\left(\sum_x{}'' \; |x\rangle + \sum_x{}' \; |x\rangle\right) = \frac{1}{\sqrt{N}}\left(\sqrt{N-M}\,|\alpha\rangle + \sqrt{M}\,|\beta\rangle\right)$$

$$= \sqrt{\frac{N-M}{N}} \; |\alpha\rangle + \sqrt{\frac{M}{N}} \; |\beta\rangle$$

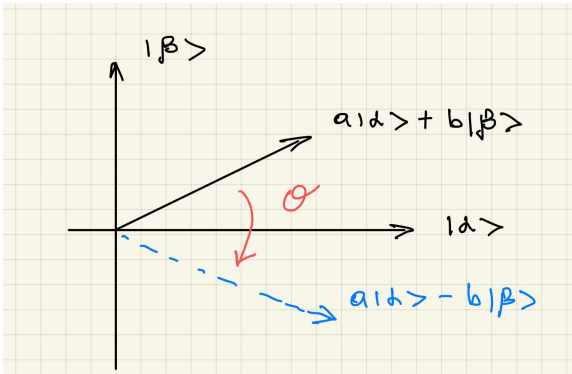
- Consider any state in span$\{|\alpha\rangle\,, |\beta\rangle\}$ with real coefficients a, b:

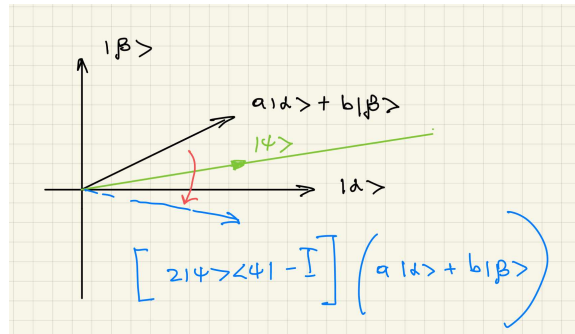
$$a\,|\alpha\rangle + b\,|\beta\rangle \;\xrightarrow{\mathcal{O}}\; a\,|\alpha\rangle - b\,|\beta\rangle = \left(2 \cdot |\alpha\rangle\,\langle\alpha| - I\right) \cdot (a\,|\alpha\rangle + b\,|\beta\rangle)$$

i.e.,

$$\mathcal{O} \equiv 2 \cdot |\alpha\rangle\,\langle\alpha| - I$$

- Note that for any vector in the span$\{|\alpha\rangle, |\beta\rangle\}$, $\mathcal{O}$ represents a reflection about $|\alpha\rangle$ vector, and likewise, $(2|\psi\rangle\langle\psi| - I)$ represents a reflection about $|\psi\rangle$ vector:

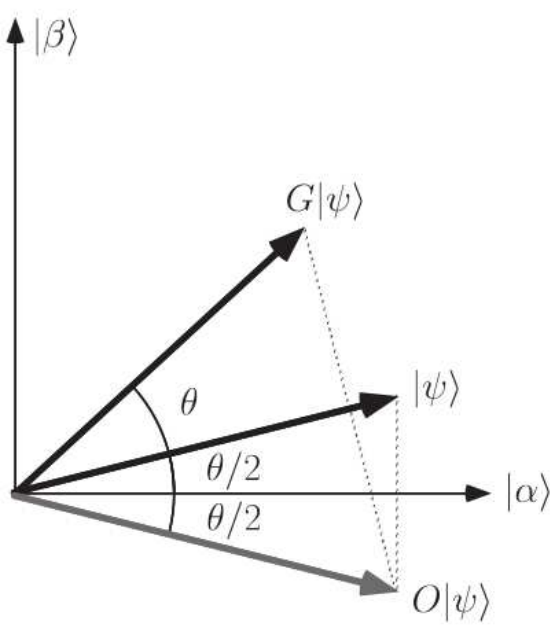




- Let's rewrite

$$|\psi\rangle = \cos\frac{\theta}{2}\ |\alpha\rangle + \sin\frac{\theta}{2}\ |\beta\rangle\ , \qquad \cos\frac{\theta}{2} = \sqrt{\frac{N-M}{N}}\,, \qquad \sin\frac{\theta}{2} = \sqrt{\frac{M}{N}}$$

$\Longrightarrow$



$$G\,|\psi\rangle = \underbrace{(2\,|\psi\rangle\langle\psi| - I)}_{\text{rotation by } +2\theta} \cdot \underbrace{(2\cdot|\alpha\rangle\langle\alpha| - I)}_{\text{rotation by } -\theta/2} \cdot |\psi\rangle = \cos\frac{3\theta}{2}\ |\alpha\rangle + \sin\frac{3\theta}{2}\ |\beta\rangle$$

$\Longrightarrow$ after k-applications of G we have:

$$G^k \left|\psi\right\rangle = \cos\left(\frac{2k+1}{2}\theta\right) \ \left|\alpha\right\rangle + \sin\left(\frac{2k+1}{2}\theta\right) \ \left|\beta\right\rangle$$

- Starting from the initial state $\left|\psi\right\rangle$ we need to rotate it as close as possible to $\frac{\pi}{2}$

$\Longrightarrow$ the total rotation angle must be

$$\phi_{rot} = \frac{\pi}{2} - \frac{\theta}{2} \implies \qquad \cos\phi_{rot} = \sin\frac{\theta}{2} = \sqrt{\frac{M}{N}} \implies \qquad \phi_{rot} = \arccos\sqrt{M/N}$$

$\Longrightarrow$ the number of rotations needed (each time G is applied we rotate by θ)

$$R = CI\left[\frac{\phi_{rot}}{\theta}\right] = CI\left[\frac{\arccos\sqrt{M/N}}{\theta}\right]$$

$\Longrightarrow$ When $M \ll N$,

$$\sin\frac{\theta}{2} \sim \frac{\theta}{2} = \sqrt{\frac{M}{N}}\,, \qquad \arccos\sqrt{M/N} \sim \frac{\pi}{2}$$

$\Longrightarrow$

$$R = CI\left[\frac{\pi}{4}\cdot\sqrt{\frac{N}{M}}\right]$$

- After R rotations, the angle between $\left|\beta\right\rangle$ and $G^k\left|\psi\right\rangle$ — we call it γ — can not be larger than $\frac{\theta}{2} \Longrightarrow$

$$G^R \left|\psi\right\rangle = \cos\gamma \ \left|\beta\right\rangle + \sin\gamma \ \left|\alpha\right\rangle$$

$\Longrightarrow$ measurement in the computational basis would produce

$$\boxed{} \left(\cos\gamma \ \left|\beta\right\rangle + \sin\gamma \ \left|\alpha\right\rangle\right) = \begin{cases} \text{one of the } \ x' \text{ states}\,, & \text{prob} = \cos^2\gamma \\ \text{one of the } \ x'' \text{ states}\,, & \text{prob} = \sin^2\gamma \end{cases}$$

$\Longrightarrow$ probability of error is

$$\sin^2\gamma \le \gamma^2 \le \left(\frac{\theta}{2}\right)^2 \ \sim \ \frac{M}{N} \ \ll 1$$